A-P-P-A-I Agent: Application Chat Flow & UI Sequence Specification

User Interaction Architecture: Action-Event Triggers, System Execution, and Screen Transitions

Executive Overview: This document maps the step-by-step user interaction flow, button click events, system state transformations, and screen navigation rules for the **Autonomous Personal Procurement Assistant Agent (A-P-P-A-I)** Chat Interface. Built on the three core MVP technical pillars—**Catalog Discovery**, **Tokenized Payment Execution**, and **Governance Guardrails**—it defines exactly what triggers when a user interacts with the system and which page comes next.

1. High-Level Screen Flow & UI State Map

Step #	Active Screen / View	User Action / Click Event	System Execution Backend	Next Destination Screen
1	Natural-Language Chat Interface (/chat)	User clicks 'Submit Query' or presses Enter with prompt	NLP Intent Parser extracts specs, price cap, and delivery date. Ingests MCP / JSON-LD feeds.	Catalog Match & Candidate Summary (/match)
2	Catalog Match & Summary View (/match)	User clicks 'Confirm Order' or Auto-Buy rule engages	Governance Engine executes hard budget pre-check (item cap & daily spend cap).	Path A: Order Confirmation (/confirmation) Path B: HITL Approval (/hitl-approval)
3A	Governance Pre-Check Passed (Path A)	Automated transition after successful pre-check	Requests PayPal Agent Ready scoped token and submits native API checkout payload.	Order Confirmation Screen (/confirmation)
3B	Human-in-the-Loop Approval Modal (/hitl-approval)	User clicks 'Approve Exception' or 'Reject Order'	If Approved: Issues token & executes checkout. If Rejected: Releases spend lock & terminates.	Approve -> Order Confirmation (/confirmation) Reject -> Chat Interface (/chat)
4	Order Confirmation & Audit Screen (/confirmation)	User clicks 'Return to Chat' or 'View Audit Trail'	Persists transaction record, updates daily spend counter, logs PayPal token hash.	Active Chat Screen (/chat) or Settings (/settings)

2. Detailed Screen Navigation & Trigger Specifications

2.1 Screen 1: Natural-Language Chat Interface (/chat)

Current Page Description: The primary conversational dashboard where the user enters natural-language procurement prompts (e.g., *'Order a replacement wireless mouse under $50 delivered by Friday'*).
1. What happens when someone clicks 'Submit Query' / 'Send Prompt':
• The UI transitions the chat state to PROCESSING_INTENT with an animated progress indicator.
• The client posts the payload to POST /api/v1/discovery/parse-intent. The NLP Intent Parser parses the prompt into structured JSON parameters: item_keyword: 'wireless mouse', max_price_constraint: 50.00, and delivery_deadline: '2026-10-30'.
• The Discovery Engine immediately calls POST /api/v1/discovery/query-catalog to search merchant Model Context Protocol (MCP) endpoints and structured JSON-LD product feeds.
• Real-time stock status, unit pricing, and delivery estimates are verified.
2. Which page comes next:
• **Catalog Match & Product Summary Screen (/match)**: Displays the retrieved product candidate, verified price, merchant details, and delivery date.

2.2 Screen 2: Catalog Match & Candidate Review Screen (/match)

Current Page Description: Card-based UI showing matched product specifications (e.g., Ergonomic Optical Replacement Mouse at $34.99), merchant rating, and total price breakdown.
1. What happens when someone clicks 'Confirm Order' / 'Proceed to Checkout':
• The button enters a loading lock state (EVALUATING_GOVERNANCE) to prevent double submissions.
• The client submits POST /api/v1/governance/evaluate-order containing user_id and proposed_item_price.
• The Governance Engine evaluates two deterministic hard budget rules within sub-second latency (≤100 ms):
- **Check 1 (Max Item Price Cap):** Verifies if proposed_item_price ≤ max_item_price_cap ($50.00).
- **Check 2 (Daily Spend Cap):** Verifies if current_daily_spend + proposed_item_price ≤ daily_spend_cap ($200.00).
2. Which page comes next:
• **Path A (Both Checks PASS):** Automatically transitions to **Token Generation & Order Execution**, then routes to **Order Confirmation Screen (/confirmation)**.
• **Path B (Threshold Breach / Exception):** Triggers a GOVERNANCE_HOLD and routes immediately to **Human-in-the-Loop Approval Screen (/hitl-approval)**.

2.3 Screen 3: Human-in-the-Loop Approval Escalation Modal (/hitl-approval)

Current Page Description: High-priority safety modal/screen displaying exact breach details (e.g., *'Proposed price $65.00 exceeds maximum item cap of $50.00'*) with explicit approval controls.
1. What happens when someone clicks 'Approve Exception' or 'Reject Order':
• **Option A — Clicking 'Approve Exception':**
- State transitions from GOVERNANCE_HOLD to APPROVED_FOR_TOKENIZATION.
- Payment module invokes PayPal Agent Ready API (POST /api/v1/payment/request-token) to issue a single-use scoped token.
- Direct API checkout executes via POST /api/v1/checkout/execute-order submitting token and shipping reference directly to merchant native API.
• **Option B — Clicking 'Reject Order' (or 15-minute Timeout):**
- State transitions to CANCELLED_BY_USER or EXPIRED_HOLD.
- Pending spend locks are immediately released, no token is requested, and execution terminates.
2. Which page comes next:
• **If Approved:** Routes directly to **Order Confirmation Screen (/confirmation)**.
• **If Rejected / Cancelled:** Routes back to **Active Chat Interface (/chat)** with cancellation logged.

2.4 Screen 4: Order Confirmation & Audit Log Screen (/confirmation)

Current Page Description: Receipt and audit dashboard displaying complete order details, merchant confirmation code, and updated governance spend counters.
1. What happens when someone clicks 'Return to Chat' or 'View Audit Trail':
• The system persists transaction logs to transaction_logs table with anonymized PayPal token reference (payment_token_ref) and merchant order ID.
• Real-time daily spend accumulator (current_daily_spend) is permanently updated.
• Clicking **'Return to Chat'** clears active intent state and resets prompt box.
• Clicking **'View Audit Trail'** opens complete security log proving zero raw credit card data was stored or transmitted.
2. Which page comes next:
• **Primary Route: Active Chat Interface (/chat)** ready for next procurement request.
• **Secondary Route: Governance & Security Settings (/settings)** to adjust caps or view logs.

3. System State Machine & Endpoint Mapping

State Name	Triggering Event / API Endpoint	Security & Governance Action	Target UI Output
PARSED	POST /api/v1/discovery/parse-intent	Extracts structured price/date constraints from prompt	Shows parsed parameters
MATCHED	POST /api/v1/discovery/query-catalog	Verifies stock & price via MCP / JSON-LD feeds	Renders product match card
GOVERNANCE_HOLD	POST /api/v1/governance/evaluate-order	Hard spend cap breach detected; locks budget delta	Launches HITL Modal

State Name	Triggering Event / API Endpoint	Security & Governance Action	Target UI Output
APPROVED_FOR_TOKENIZATION	POST /api/v1/payment/request-token	Generates single-use PayPal Agent Ready token	Triggers checkout payload
SUCCESS / EXECUTED	POST /api/v1/checkout/execute-order	Merchant native checkout API executes; logs audit	Order Confirmation Screen